

Azure SDK Authentication and Azure CLI Login

When do Azure, OpenAI, and Microsoft SDKs require az login?

Short Answer

No. Using an Azure SDK, the OpenAI SDK with Azure resources, or another Microsoft SDK does not automatically mean that you must be logged in with the Azure CLI. Whether **az login** is required depends on the authentication method your code is using.

Core rule: The SDK needs credentials. Azure CLI login is only one possible source of those credentials.

DefaultAzureCredential

A common Azure SDK pattern is to use DefaultAzureCredential. It tries a chain of supported credential sources. During local development, one of those sources can be your existing Azure CLI login.

```
from azure.identity import DefaultAzureCredential
```

```
credential = DefaultAzureCredential()
```

If your local development environment relies on Azure CLI as the credential source, you would first run:

```
az login
```

In that case, the SDK can reuse the Microsoft Entra identity established by the CLI session. However, DefaultAzureCredential can also succeed through other supported credentials, so Azure CLI is not inherently required.

Authentication Methods and Whether az login Is Required

Authentication method	az login required?	Typical use
AzureCliCredential()	Yes	Local development
DefaultAzureCredential() using Azure CLI	Yes	Local development
DefaultAzureCredential() using another supported credential	No	Local development
Managed Identity	No	Azure Functions, App Service, VMs, Container Apps, etc.
Service Principal / App Registration	No	Applications, automation, CI/CD
Workload Identity	No	Kubernetes and CI/CD
API key	No	Services that support key authentication
Connection string / account key	No	Some Azure services

When Azure CLI Login Is Explicitly Required

If your code explicitly uses AzureCliCredential, Azure CLI is the credential provider:

```
from azure.identity import AzureCliCredential

credential = AzureCliCredential()
```

In this case, Azure CLI must be installed and your CLI session must be authenticated with a valid login.

Azure OpenAI and Microsoft Foundry Example

Azure OpenAI and Microsoft Foundry scenarios can use different authentication approaches, depending on the service and configuration.

API key authentication

```
client = AzureOpenAI(
    api_key=os.environ["AZURE_OPENAI_API_KEY"],
    azure_endpoint=endpoint,
    api_version="..."
)
```

With API-key authentication, you generally do not need to run az login because the API key authenticates the request.

Microsoft Entra ID authentication

```
from azure.identity import DefaultAzureCredential, get_bearer_token_provider

credential = DefaultAzureCredential()

token_provider = get_bearer_token_provider(
    credential,
    "https://cognitiveservices.azure.com/.default"
)
```

Here the application uses Microsoft Entra ID. During local development, DefaultAzureCredential may obtain your identity from Azure CLI. When the same application is deployed to Azure, a managed identity can be used instead.

Typical Local Development Flow

LOCAL DEVELOPMENT

Your Python / backend application

|

v

DefaultAzureCredential()

|

+--> Azure CLI session

az login

|

v

Microsoft Entra ID

|

v

Azure / Foundry resource

Typical Azure Production Flow

AZURE PRODUCTION

Azure Function / App Service / Container App

|

v

Managed Identity

|

v

Microsoft Entra ID

|

v

Azure AI Search

Azure OpenAI / Foundry

Blob Storage

Other Azure resources

There is no interactive az login in this production flow. Azure provides the application's managed identity.

Authentication vs. Authorization

Authentication answers: Who are you?

Authorization answers: What are you allowed to do?

Successfully running az login proves your identity, but it does not automatically grant access to every Azure resource. The authenticated identity still needs the appropriate Azure RBAC role for the operation being performed.

Practical Rules of Thumb

- **AzureCliCredential()** - Expect to run az login.
- **DefaultAzureCredential()** - az login is often used during local development, but another supported credential may satisfy it.
- **API key** - You generally do not need az login.
- **Managed Identity in Azure** - You do not run az login; Azure supplies the application's identity.

This is why some Microsoft Learn or Foundry notebook examples have you run az login even though the SDK itself is not inherently dependent on Azure CLI: the lab is using Azure CLI as a convenient local credential source.

Microsoft Documentation References

- Azure Identity - DefaultAzureCredential (Python):
<https://learn.microsoft.com/en-us/python/api/azure-identity/azure.identity.defaultazurecredential>
- Azure SDK authentication for local development:
<https://learn.microsoft.com/en-us/azure/developer/python/sdk/authentication/local-development-dev-accounts>
- Keyless connections for Azure AI services:
<https://learn.microsoft.com/en-us/azure/developer/ai/keyless-connections>
- Microsoft Foundry / Azure OpenAI Entra ID configuration:
<https://learn.microsoft.com/en-us/azure/foundry/foundry-models/how-to/configure-entra-id>